Finite Groups and Their Representations — Exercise Sheet 11

Local moves and the Jucys–Murphy spectrum

Variant: Questions only

Exercise 1 (The local relations without a positive inner product). Let $a, b \in \mathbb{C}$ with $a \neq b$ and put

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \qquad B = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}.$$

Find every matrix S satisfying $S^2 = \mathbb{1}$ and $SA + \mathbb{1} = BS$. Include the twin relation $AS + \mathbb{1} = SB$ in your verification. Classify these modules under changes of basis preserving A and B. Here a subspace is invariant if it is stable under A, B, S. Decide when the module is irreducible, a direct sum of invariant lines, or reducible but not a direct sum. Explain why the last case admits no positive-definite inner product making all three matrices self-adjoint.

Exercise 2 (A local move with $r = -3$). Equip an irreducible S_6-module with an invariant inner product. Suppose it contains a unit GZ vector v of weight $\alpha = (0, 1, 2, -1, 0, 1)$. For $i = 3$, put $r = a_4 - a_3$ and compute $w = s_3 v - r^{-1} v$, its weight and squared norm. Give the matrices of s_3, X_3, X_4 on (v, w) and the Gram matrix in this basis. Then give the matrix of s_3 on the corresponding orthonormal basis. This is a conditional application of the local theorem; no converse to $\mathrm{Spec}(6) \subseteq \mathrm{Cont}(6)$ is assumed.

Exercise 3 (Reconstructing a tableau or finding the obstruction). For each vector below, decide whether it satisfies the intrinsic content conditions of Lecture 11. If it does, reconstruct its standard tableau. If it does not, identify a failed condition and its positions.

$$\alpha^{(1)} = (0, 1, 2, -1, 0, 1), \quad \beta^{(1)} = (0, 1, 2, -1, 1, 0),$$
$$\beta^{(2)} = (0, 1, 2, -1, 0, 4), \quad \alpha^{(6)} = (0, -1, -2, 1, 0, -1).$$

Do not infer existence of a representation from the combinatorial bijection. Also suppose, for a contradiction, that $\beta^{(1)}$ is a GZ weight. Apply the admissible exchange s_4 and use the braid relation to exclude it.

Exercise 4 (The graph of admissible moves). Start with $\alpha^{(1)} = (0, 1, 2, -1, 0, 1)$. Join two content vectors by an edge labelled i when they differ by the admissible exchange s_i of entries $i, i + 1$. Find the whole connected component of $\alpha^{(1)}$ and all its labelled edges. Reconstruct the five tableaux and show that they have shape $(3, 3)$. Explain what the local theorem implies about these vectors if any one is a GZ weight of an irreducible module.

Exercise 5 (The transposition sum on a weight). Let v be a GZ vector of weight $\alpha = (a_1, \ldots, a_n)$ in an irreducible module U of S_n. Prove that τ_n acts on v by $\sum_k a_k$. For a standard tableau of shape λ, express the sum of its contents in terms of the row lengths of λ and the column lengths $\lambda'_j = \#\{i : \lambda_i \geq j\}$, the parts of its conjugate partition λ'. If $n = 6$ and $\alpha = (0, 1, 2, -1, 0, 1)$, find $\chi_U((1\,2))/\dim U$. State separately what happens at $n = 1$.

Exercise 6 (Symmetric polynomials and central characters). Use the following theorem as a given: the centre of $\mathbb{C}[S_n]$ is exactly the set of symmetric polynomials in $X_1, \ldots, X_n$

(Jucys–Murphy theorem, Ceccherini-Silberstein–Scarabotti–Tolli, §4.4). Prove that two GZ weights give the same central character exactly when their unordered multisets of entries agree. Use the elementary symmetric polynomials and $\prod_k (t - a_k)$ for the converse. Here the elementary symmetric polynomials are

$$e_0 = 1, \qquad e_k(x_1, \dots, x_n) = \sum_{1 \le i_1 < \cdots < i_k \le n} x_{i_1} \cdots x_{i_k} \quad (1 \le k \le n).$$

Expanding the product gives Viète's formulas

$$\prod_{j=1}^{n} (t - a_j) = \sum_{k=0}^{n} (-1)^k e_k(a_1, \dots, a_n) t^{n-k}.$$

Compare the following vectors, conditionally on their occurring as weights:

$$\alpha^{(1)} = (0, 1, 2, -1, 0, 1), \qquad \alpha^{(2)} = (0, 1, -1, 2, 0, 1),$$

$$\alpha^{(7)} = (0, 1, 2, 3, -1, -2).$$

Show why the single central element τ_6 does not settle the second comparison, and exhibit a symmetric polynomial that does.

Exercise 7 (⋆ All two-dimensional local modules). Classify up to simultaneous change of basis all triples of complex 2×2 matrices (A, B, S) satisfying

$$AB = BA, \qquad S^2 = \mathbb{1}, \qquad SA + \mathbb{1} = BS.$$

There is no assumption that A or B is diagonalisable. Put $C = A + B$ and $D = B - A$. Prove that C commutes with S and D and $SD + DS = 2\mathbb{1}$. Split into the cases $S = \pm\mathbb{1}$ and $S \sim \operatorname{diag}(1, -1)$. Distinguish split modules, reducible non-split modules and irreducible modules. Give all canonical families and the parameters distinguishing their isomorphism classes, including cases where A and B are not diagonalisable. This extension is estimated at 45 minutes.